Knowledge-Grounded Reasoning and Self-Improvement for Autonomous AI Agents

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Overview

Autonomous agents must learn unfamiliar facts, plan multi-step investigations, use tools, and revise conclusions as new observations arrive. Language models may complete a task with invalid intermediate steps or apply a correct rule under the wrong conditions. Training on final outcomes alone provides limited information about which reasoning behaviors are trustworthy and transferable.

Central research question: How can agents acquire unfamiliar external knowledge, learn checkable reasoning processes, and autonomously verify and revise conclusions?

We propose three connected research thrusts:

- **Structured knowledge for open-world reasoning.** We will retrieve and extract entities, events, relations, scientific principles, and contextual constraints from heterogeneous sources. Compact evidence records will distinguish observations, inferred relations, unresolved assumptions, and the conditions under which a scientific or logical rule applies.
- **Scalable process-guided learning.** We will synthesize multi-hop reasoning tasks and intermediate trajectories from structured knowledge, including controlled reasoning errors. Fine-grained evaluators will identify factual, contextual, and logical failures. Process rewards will guide candidate-step selection and iterative training; we will test transfer to held-out knowledge compositions.
- **Verification-driven autonomous reasoning.** Agents will form hypotheses or plans, expose required assumptions, select evidence-retrieval and symbolic or executable tools, and use diagnostic feedback to repair unsupported steps. Scientific hypothesis discovery, spanning literature retrieval and executable validation, will provide a challenging testbed for open-world generalization.

Evaluation. Evaluation will measure reasoning-step validity, outcome accuracy, error localization, repair success, transfer across conditions, and computational cost. Baselines will include retrieval-augmented agents, outcome-only learning, conventional process reward models, and tool-using agents without explicit condition tracking.

Naval relevance and potential impact. The project develops foundational methods for intelligent autonomous agents capable of reasoning under incomplete information and checking why a proposed action or conclusion is justified, with potential long-term relevance to Naval autonomy and planning.